INCH-POUND
MIL-DTL-16377/72A(SH)
30 June 2014
SUPERSEDING
MIL-DTL-16377/72(SH)
02 August 1996

DETAIL SPECIFICATION SHEET

FIXTURES, LIGHTING; BALLAST, FOR ONE 20-WATT T12 FLUORESCENT LAMP

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-16377.

All dimensions are in inches. For tolerances, other than specified herein, see 3.5.2 of MIL-DTL-16377.

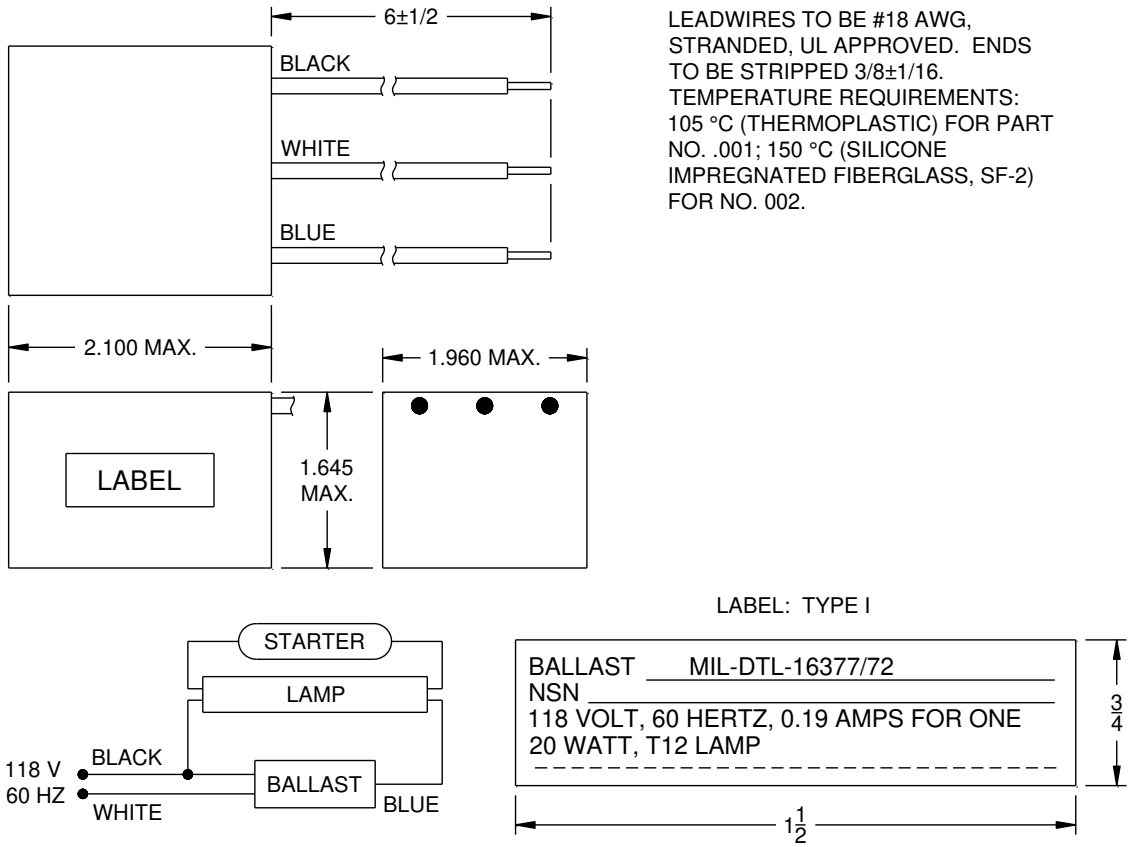


FIGURE 1. Ballast for one 20-watt T12 fluorescent lamp.

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REQUIREMENTS:

DESIGN:

Dimension and configurations:	See figure 1.
Assembly:	The ballast shall consist of a laminated iron core and coil, integrally assembled and conforming to the dimensions and requirements shown herein.
Impregnating:	Ballast circuitry to be housed in thermoset plastic potting shell and to be potted with epoxy.
Markings:	Ballast shall conform to marking requirements and color coding of leads as specified in the applicable paragraph of NEMA ANSI C82.1.
Label:	Label shall be in accordance with Type I except as specified herein.

EXAMINATION AND TESTS:

Examination and tests shall be as specified in MIL-DTL-16377 and as follows:

First article inspection: See table II.

Conformance:

Comparison inspection: Same as first article inspection.

Conformance inspection:

a. Power output (except ratio peak current to r.m.s. current).
b. Dielectric withstanding voltage.
c. Insulation resistance.
d. Voltage drop across ballast.
e. Current harmonic distortion.
f. Ballast efficiency.
g. Ballast power factor.

GENERAL INFORMATION:

Assignment of military part numbers and general information is shown in table I.

TABLE I. Ballast rating and general information.

Military Part No. M16377/72-	Rating				Efficiency	Ballast Factor	National Stock Number (NSN)	Remarks
	Volt	Amperes	Hertz	Power Factor				
001	118	0.19	60	≥90%	>80%	92.5%	6250-01-465-6716	Standard
002	118	0.19	60	≥90%	>80%	92.5%	6210-01-510-9785	Heat resistant

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TABLE II. First article and comparison inspection. [1/]

Test	Requirement	Test Method	Remarks
Examination	3.13	4.5	[2/]
Performance a. Power output b. Regulation c. Noise d. Voltage drop across ballast e. Ballast efficiency f. Ballast power factor	[3/]	[4/]	[2/]
Dielectric withstanding voltage	3.6.15	4.8.2	[2/]
Insulation resistance	3.6.16	4.8.3	[2/]
Current harmonic distortion	3.6.18.1	4.8.22	[2/]

NOTES:

[1/] Requirements and test method paragraph numbers refer to MIL-DTL-16377.

[2/] Sample lot size for inspection shall consist of five ballasts.

[3/] The performance of the ballasts shall be in conformance with the requirements listed in the applicable paragraph of NEMA ANSI C82.1, except that the voltage drop across the ballast shall be not less than 84 volts.

[4/] Test procedure for performance shall be as follows:

a. Performance tests shall be made in accordance with NEMA ANSI C82.2.
b. Fluorescent lamp reference ballasts shall be in conformance with NEMA ANSI C82.3.
c. Lamp shall conform to ANSI/ANSLG C78.81.
d. Voltage across starter terminals and across ballast, when supplying a reference lamp, shall be measured.

Reference ballast characteristics: The specifications described in table III ensure that the ballasts conform to requirements to operate the F20T12CW lamp, PPD 802-6215499.

TABLE III. Reference ballast characteristics for field light output. [1/]

Input Voltage	Lamp Current	Impedance	Power Factor	Volt-Amps
118V	0.327A	276.9 ohms	0.071	29.6

NOTE:

[1/] Test conditions (77 °F [25 °C] ambient).

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CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

Custodian:
Navy – SH

Review activity:
DLA – IS

Preparing activity:
Navy – SH
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NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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